Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

| | | | | |
|---|---|---|---|---|
| Title of Proposed Technology | A MoSiB(Ti) next generation superalloy for turbine applications | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (x ), Hydrogen ( x), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | **Mechanical・Materials** | | **Mechanical・Materials** | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Aerospace Systems | Fixed-wing/Rotary-wing Aircraft Propulsion Systems | Metal Materials | Structural Materials |
| Concept of Proposed Technology | Nickel-based superalloys such as Inconel have represented the industrial standard for high-temperature applications in aircraft and gas turbines for decades. However, their operational limits are reached at approximately 1,000 °C, which constrains further efficiency improvements in modern turbine concepts, particularly in the context of hydrogen-based combustion systems. Future aerospace and energy technologies require materials that combine higher maximum service temperatures, significantly reduced creep rates, improved oxidation resistance, and lower density while maintaining or exceeding current strength levels.<br>The Mo–Si–B-(Ti) material system represents a highly promising candidate to meet these requirements. In particular, additive manufacturing via Electron Beam Powder Bed Fusion (PBF-EB) has enabled the realization of exceptionally fine microstructures and outstanding mechanical properties at Fraunhofer IFAM Dresden. Four-point bending strengths of approximately 1,200 MPa | | | |

at 1,000 °C have already been achieved, representing a substantial improvement over conventionally powder-metallurgical Mo–Si–B materials.

Targeted alloying with titanium offers additional potential to further enhance creep resistance while simultaneously improving oxidation behavior. To date, however, the additive manufacturing of Mo–Si–B–Ti alloys has not yet been demonstrated worldwide. This is precisely where the proposed German–Korean joint project is positioned: the systematic development, characterization, and validation of additively manufactured Mo–Si–B and Mo–Si–B–Ti materials as next-generation high-temperature turbine materials.

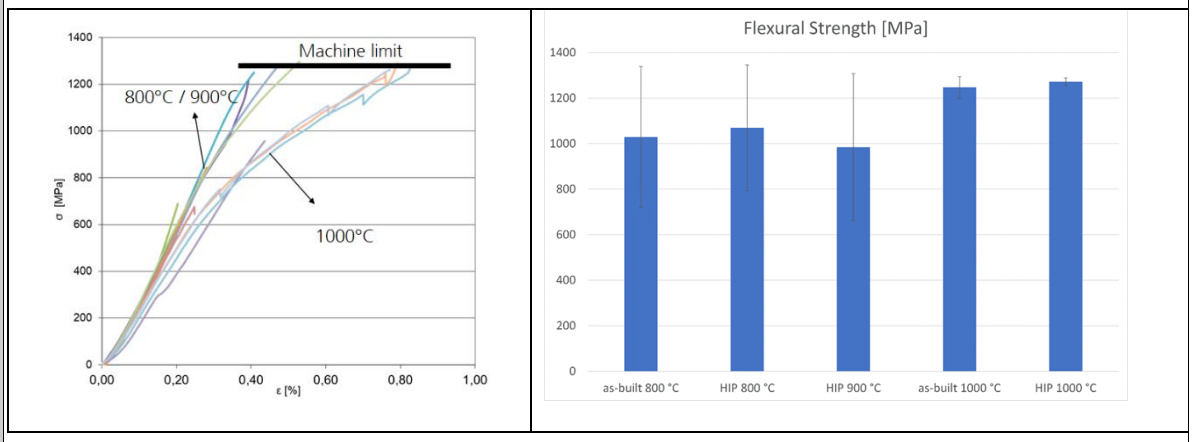


**Figure1**: Stress - strain graph (left) and bending strength (right) obtained at 4-point bending strength of Mo9Si8B (at. %) processed by PBF-EB and HIP [10]

**Rationale for Joint R&D Project**

Achieving these ambitious development goals requires close international collaboration between a leading research institute and an industrially oriented SME. Fraunhofer IFAM Dresden provides well-established expertise in powder metallurgy, additive manufacturing of refractory metals, and advanced microstructural characterization. However, dedicated facilities for high-temperature creep testing in the relevant temperature and stress regimes are currently not available in-house.

A Korean SME with expertise in high-temperature mechanical testing, alternative manufacturing technologies (e.g., LMD, SLM, or powder metallurgy), or powder production can ideally complement these capabilities. At the same time, the SME gains early access to a disruptive material technology with high commercialization potential in aerospace, power generation, and hydrogen turbine applications. The collaboration allows risks to be shared, development timelines to be shortened, and a clear pathway toward later industrial implementation to be established.

**Key Words**

1. High temperature materials
2. Creep resistance
3. Additive manufacturing
4. Next generation superalloys
5. MoSiB

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 3 ) | End of the Project | Level ( 4 ) |
|---|---|---|---|---|
| **R&D Objectives** | The objective of the project is the development and validation of additively manufactured Mo–Si–B and Mo–Si–B–Ti materials with significantly improved high-temperature performance compared to state-of-the-art nickel-based superalloys. Specifically, the project aims to:<br><br>• establish process–property relationships for PBF-EB or alternative manufacturing technologies,<br>• determine, for the creep properties of additively manufactured Mo–Si–B materials in the temperature range of 1,000–1,300 °C,<br>• demonstrate and evaluate the additive processing of Mo–Si–B–Ti alloys,<br>• create the foundation for industrial scalability and future demonstrator components. | | | |
| **R&D Target Specifications** | • Maximum service temperature > 1,100 °C<br>• Creep strain rate at least one order of magnitude lower than that of Ni-based superalloys under comparable temperature and stress conditions<br>• High oxidation resistance under relevant turbine atmospheres<br>• Density significantly lower than that of conventional Ni-based superalloys | | | |
| **R&D Plan** | **Phase 1: Pre-Project (6 months) – Feasibility and Risk Mitigation**<br><br>WP 1.1 Material and Process Screening<br>• Procurement of gas-atomized Mo–Si–B powder<br>• Initial additive manufacturing trials (PBF-EB, optionally alternative processes provided by the Korean partner)<br>• Assessment of processability, crack susceptibility, and resulting microstructures<br><br>WP 1.2 Preliminary Material Characterization<br>• Density measurements, phase analysis (XRD), microstructural characterization (SEM/EDS)<br>• Identification of promising process windows<br><br>WP 1.3 Phase 2 Project Definition<br>• Selection of the most suitable manufacturing route<br>• Definition of alloy compositions (including Ti content)<br>• Detailed planning of creep and high-temperature mechanical testing<br><br>**Outcome of Phase 1:** Robust feasibility assessment, clearly defined development roadmap for the main project, and decision basis for both funding agencies.<br><br>**Phase 2: Main Project (30 months) – Development, Validation, and Demonstration**<br><br>WP 2 Material Development and Additive Manufacturing<br>• Optimization of AM parameters for Mo–Si–B and Mo–Si–B–Ti<br>• Heat treatment and HIP for microstructure control<br>• Establishment of a process and material data repository (Product Information Table, PIT)<br><br>WP 3 High-Temperature Mechanical Testing<br>• Creep testing (1,000–1,300 °C, 100–500 MPa)<br>• Complementary fatigue and fracture toughness testing<br>• Benchmarking against Ni-based superalloys | | | |

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WP 4 Microstructural and Mechanism Analysis

- SEM, EBSD, TEM investigations and phase stability analysis
- Identification of dominant creep and damage mechanisms
- Feedback loop for further material and process optimization

WP 5 Demonstrators and Exploitation Preparation

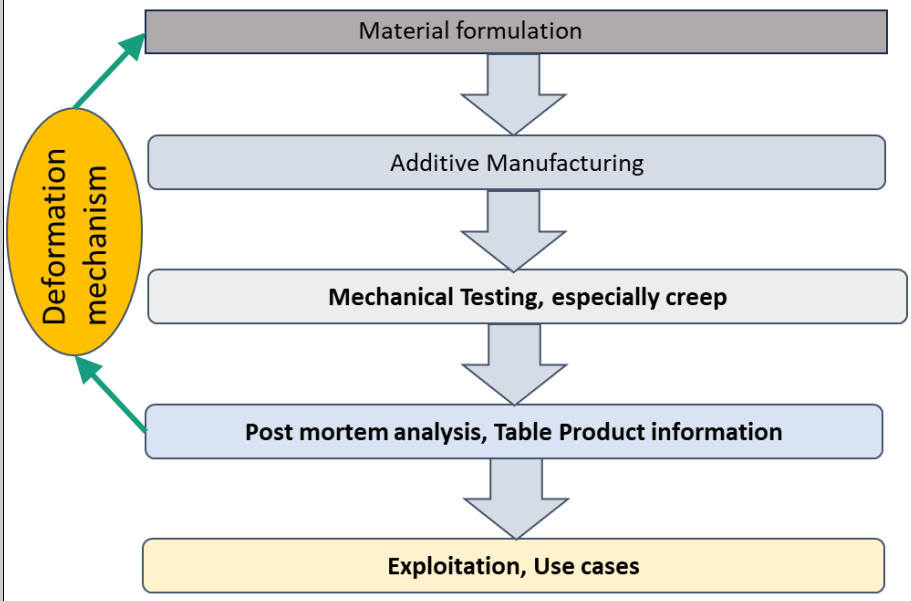
- Manufacturing of representative demonstrator components
- Assessment of industrial scalability
- Preparation of IP, technology transfer, and exploitation strategies

WP 6 Dissemination and Network Building

- Joint scientific publications and conference contributions
- Engagement with additional industrial stakeholders
- Visibility at trade fairs and technical events

**Milestones**

**MS 1 (Month 6):** Successful completion of Phase 1 and definition of optimized process route
**MS 2 (Month 18):** Validated creep data for additively manufactured Mo–Si–B
**MS 3 (Month 30):** Demonstration of additive manufacturing of Mo–Si–B–Ti alloys
**MS 4 (Month 36):** Demonstrator component and defined exploitation strategy



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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | Fraunhofer IFAM Dresden will act as the scientific and technological lead of the project, focusing on materials development, additive manufacturing, and fundamental understanding of structure–property relationships. The institute will contribute its long-standing expertise in refractory metal systems, powder metallurgy, and advanced additive manufacturing processes.<br><br>Key responsibilities of IFAM include:<br><br>• Definition and optimization of alloy concepts within the Mo–Si–B and Mo–Si–B–Ti material systems, including composition design and powder specifications.<br>• Additive manufacturing of test specimens and demonstrator components using PBF-EB, including parameter development for crack-free processing and microstructure refinement.<br>• Post-processing of additively manufactured components, including heat treatment and hot isostatic pressing (HIP), to achieve fully dense materials and tailored microstructures.<br>• Comprehensive microstructural and phase analysis (SEM, EDS, EBSD, XRD) before and after mechanical testing, enabling a fundamental understanding of deformation and damage mechanisms.<br>• Development and maintenance of a structured process–property database (Product Information Table, PIT) linking manufacturing parameters, microstructure, and mechanical performance.<br>• Scientific coordination of the project, including feedback loops between manufacturing, testing, and analysis, as well as joint publication and dissemination activities.<br><br>Through these activities, IFAM ensures the scientific excellence of the project and provides the knowledge base required for systematic material optimization and future technology transfer. |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will play a key role in bridging the gap between laboratory-scale material development and industrial applicability. The SME is expected to contribute strong capabilities in high-temperature testing, manufacturing technologies, and application-oriented evaluation, thereby ensuring industrial relevance from an early project stage.<br><br>Key responsibilities of the Korean SME can include:<br>• Execution of high-temperature mechanical testing, in particular creep testing in the temperature range of 1,000–1,300 °C under relevant stress conditions (tension or compression), which is a core enabler of the project.<br>• Contribution of complementary manufacturing technologies (e.g., SLM, LMD, powder metallurgy, or alternative consolidation routes) to benchmark and potentially scale the developed material concepts beyond PBF-EB.<br>• Advanced microstructural and chemical analysis capabilities, including TEM investigations, to support the identification of creep and oxidation mechanisms.<br>• Assessment of industrial feasibility, including manufacturability, cost considerations, and scalability toward larger components or series production.<br>• Active participation in exploitation planning, including identification of target applications, customers, and potential commercialization pathways in aerospace, energy, and hydrogen turbine markets.<br>•<br>By combining testing infrastructure, industrial perspective, and manufacturing know-how, the Korean SME ensures that the developed materials move beyond academic proof-of-concept toward practical and economically viable solutions. |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The project addresses a key bottleneck in future turbine and hydrogen energy technologies. The developed materials enable higher efficiencies, extended component lifetimes, and a significant reduction in $CO_2$ emissions. For the Korean SME, early access to a novel class of high-temperature materials creates a strong competitive advantage, while the project simultaneously delivers high-level scientific insights with a clear industrial perspective. The German–Korean collaboration will sustainably strengthen innovation capabilities in both countries in the field of advanced high-temperature and aerospace materials. |

## 6. References

[1] K. Sahithya, I. Balasundar, P. Pant, T. Raghu, Journal of Alloys and Compounds 849 (2020) 156626.
[2] D. Saber, I.S. Emam, R. Abdel-Karim, Journal of Alloys and Compounds 719 (2017) 133–141.
[3] T. Dudziak, L. Boron, A. Gupta, S. Saraf, P. Skierski, S. Seal, N. Sobczak et al., Corrosion Science 138 (2018) 326–339.
[4] J.R. Nicholls, N.J. Simms, in: R.A. Cottis (Ed.), Basic concepts, high temperature corrosion, 4th ed., Elsevier, 2010, pp. 518–540.
[5] A. Zieliński, M. Sroka, T. Dudziak, Materials (Basel, Switzerland) 11 (2018).
[6] U. Gaitzsch, A. Magnusson, T. Weißgärber, International Journal of Refractory Metals and Hard Materials special Issue "Plansee Seminar" (2023).
[7] M. Azimovna Azim, S. Burk, B. Gorr, H.-J. Christ, D. Schliephake, M. Heilmaier, R. Bornemann et al., Oxid Met 80 (2013) 231–242.
[8] D. Schliephake, M. Azim, K. von Klinski-Wetzel, B. Gorr, H.-J. Christ, H. Bei, E.P. George et al., Metall and Mat Trans A 45 (2014) 1102–1111.
[9] C. Hochmuth, D. Schliephake, R. Völkl, M. Heilmaier, U. Glatzel, Intermetallics 48 (2014) 3–9.
[10] U. Gaitzsch, A. Kirchner, A. Magnusson, T. Weißgärber, in: M. Krüger, M. Heilmaier, F. Pyczak, M. Schloffer, F. Stein (Eds.), Intermetallics 2023: International Conference, 02-06 October 2023 Educational Center Kloster Banz, Germany abstracts, 1st ed., Conventus Congressmanagement & Marketing GmbH, Jena, 2023, pp. 37–38.

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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